

This workbook has both paper and Jupyter notebook submissions.

1. Consider the image in the file "Letter_I.jpg"
 - (a) Write pseudocode and a program to take the 2D DFT of the image. Make sure that the zero frequency is in the middle of the Fourier plane. (This could take a minute or two to run on your computer.)
 - (b) Explain what you see in the Fourier plane in terms of what is present in the image.
2. A 50 fs laser pulse is modeled along a time array. We need the minimum resolution in time to 0.1 fs and in frequency 1e12 Hz. What are the required number of data points in our simulation and the length of the time array? With those values, what is the maximum frequency we can model? (*Hint*: 1 fs = 10^{-15} s.)